

Problem 1. (2019 XXII All-Ukrainian Tournament of Young Mathematicians P17) n points are marked on the board points that are vertices of the regular n -gon. One of the points is a chip. Two players take turns moving it to the other marked point and at the same time draw a segment that connects them. If two points already connected by a segment, such a move is prohibited. A player who can't make a move, loses. Which of the players can guarantee victory?

Problem 2. (2021 Ukraine NMO P9) Alexey and Bogdan play a game with two piles of stones. In the beginning, one of the piles contains 2021 stones, and the second is empty. In one move, each of the guys has to pick up an even number of stones (more than zero) from an arbitrary pile, then transfer half of the stones taken to another pile, and the other half - to remove from the game. Loses the one who cannot make a move. Who will win this game if both strive to win, and Bogdan begins?

Problem 3. (SAMO 2017 Q4) Andile and Zandre play a game on a 2017×2017 board. At the beginning, Andile declares some of the squares forbidden, meaning the nothing may be placed on such a square. After that, they take turns to place coins on the board, with Zandre placing the first coin. It is not allowed to place a coin on a forbidden square or in the same row or column where another coin has already been placed. The player who places the last coin wins the game. What is the least number of squares Andile needs to declare as forbidden at the beginning to ensure a win? (Assume that both players use an optimal strategy.)

Problem 4. Ana and Bruno decide to play a game with the following rules. Ana has cards $1, 3, 5, 7, \dots, 2n - 1$, and Bruno has cards $2, 4, 6, 8, \dots, 2n$. During the first turn and all odd turns afterwards, Bruno chooses one of his cards first and reveals it to Ana, and Ana chooses one of her cards second. Whoever's card is higher gains a point. During the second turn and all even turns afterwards, Ana chooses one of her cards first and reveals it to Bruno, and Bruno chooses one of his cards second. Similarly, whoever's card is higher gains a point. During each turn, neither player can use a card they have already used on a previous turn. The game ends when all cards have been used after n turns. Determine the highest number of points Ana can earn.

Problem 5. (2019 Central American and Caribbean Mathematical Olympiad, P2) We have a regular polygon P with 2019 vertices, and in each vertex there is a coin. Two players Azul and Rojo take turns alternately, beginning with Azul, in the following way: first, Azul chooses a triangle with vertices in P and colors its interior with blue, then Rojo selects a triangle with vertices in P and colors its interior with red, so that the triangles formed in each move don't intersect internally the previous colored triangles. They continue playing until it's not possible to choose another triangle to be colored. Then, a player wins the coin of a vertex if he colored the greater quantity of triangles incident to that vertex (if the quantities of triangles colored with blue or red incident to the vertex are the same, then no one wins that coin). The player with the greater quantity of coins wins the game. Find a winning strategy for one of the players. (Two triangles can share vertices or sides.)

Problem 6. (Kosovo TST 2019 P1) There are 2019 cards in a box. Each card has a number written on one of its sides and a letter on the other side. Amy and Ben play the following game: in the beginning Amy takes all the cards, places them on a line and then she flips as many cards as she wishes. Each time Ben touches a card he has to flip it and its neighboring cards. Ben is allowed to have as many as 2019 touches. Ben wins if all the cards are on the numbers' side, otherwise Amy wins. Determine who has a winning strategy.

Problem 7. (Brazilian Undergrad MO 2020 P5) Let N a positive integer. In a spaceship there are $2 \cdot N$ people, and each two of them are friends or foes (both relationships are symmetric). Two aliens play a game as follows:

1. The first alien chooses any person as she wishes.
2. Thenceforth, alternately, each alien chooses one person not chosen before such that the person chosen on each turn be a friend of the person chosen on the previous turn.
3. The alien that can't play in her turn loses.

Prove that second player has a winning strategy if, and only if, the $2 \cdot N$ people can be divided in N pairs in such a way that two people in the same pair are friends.

Problem 8. (Putnam 2013 B6) Let $n \geq 1$ be an odd integer. Alice and Bob play the following game, taking alternating turns, with Alice playing first. The playing area consists of n spaces, arranged in a line. Initially all spaces are empty. At each turn, a player either

1. places a stone in an empty space, or
2. removes a stone from a nonempty space s , places a stone in the nearest empty space to the left of s (if such a space exists), and places a stone in the nearest empty space to the right of s (if such a space exists).

Furthermore, a move is permitted only if the resulting position has not occurred previously in the game. A player loses if he or she is unable to move. Assuming that both players play optimally throughout the game, what moves may Alice make on her first turn?